

Designing in Flow

AI-Supported "Vibe Coding" and the Future of Instructional Design Practice

Using AI IDE Agents (Cursor & Google Antigravity) for Rapid Educational Prototyping

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"What happens when instructional designers can prototype at the speed of thought?"

1

Then

Ideas → requirements → developers → long iteration cycles

2

Now

Designers directly create and test prototypes without developer handoffs

3

Why

AI agents collapse the friction between ideation and implementation



DEFINITION

What is "Vibe Coding"?

"An AI-supported development workflow where natural language, rapid iteration, and agentic assistance allow creators to prototype software through conversational design processes."

It's not magic. It's guided orchestration.



Prompting

Natural language drives the creative process



Iteration

Rapid cycles replace lengthy revisions



Co-creation

Human orchestration, not manual coding



Rapid Prototyping

From concept to functional artifact — fast

WHY THIS MATTERS FOR IDT

From Storyboards to Living Prototypes

Traditional ID

Storyboards, static mockups, and developer handoffs drive the process

AI-Supported IDE

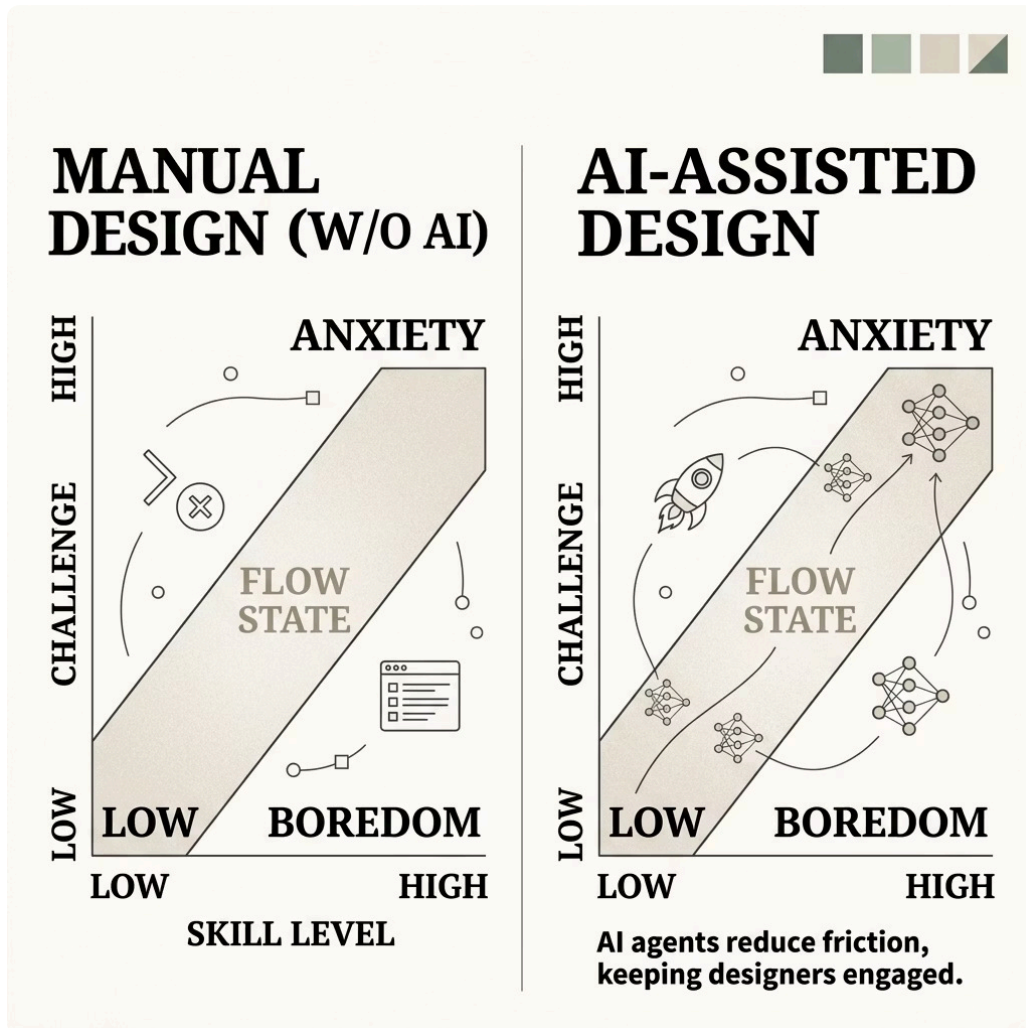
Live prototypes and functional tools built directly by the designer

Emerging Possibilities

Faster iteration, immediate learner testing, and democratized creation

Grounded in rapid prototyping, design thinking, learner-centered iteration, and experiential learning — AI-supported IDEs don't replace the designer's expertise; they amplify it.

Flow Theory & AI Design Environments



Csikszentmihalyi Applied to IDT

- Traditional coding interrupts flow for non-programmers — syntax anxiety pulls designers out of the creative zone
- AI agents absorb boilerplate and syntax, reducing "friction costs" that break concentration
- Designers stay in conceptual, pedagogical, creative flow — longer and more productively
- The human remains the **pedagogical architect**; AI handles execution details

This Workshop Is a Live Design Case

Design Decisions

Every prompt is a design choice — we examine tradeoffs in real time

Constraints & Iteration

Real tools, real limitations — iteration is the method, not the exception

Reflection-in-Action

Following Schön: we learn by doing, then pausing to analyze what happened

- 📘 Donald Schön's *Reflective Practitioner* frames this as professional learning through deliberate, situated problem-solving — exactly what AI-assisted design enables.





SECTION 2 — LIVE DEMO

"Let's Build Something Together"

We're about to prototype a real educational tool — live, in this room, in front of you.

"What educational app or tool would **actually** help your learners right now?"

Your challenge becomes the brief. We design, prompt, iterate — and reflect on every decision together.

Opportunities & Tensions

Opportunities

- Rapid prototyping compresses design-to-test cycles dramatically
- Democratized development — designers build what once required a dev team
- More experimentation, lower cost of being wrong
- Faster, richer learner feedback loops

Tensions

- Overreliance on AI may erode foundational design judgment
- Risk of pedagogical shallowness behind polished surfaces
- Ethical concerns around authorship, accessibility, and data
- Hallucinated functionality can mislead stakeholders

The Future of IDT?

"Will future designers need to code — or orchestrate AI agents?"

"Perhaps the future of IDT is not programming expertise — but **design expertise amplified by AI.**"

The Skill Shifts

From syntax to strategy. From building to orchestrating.

The Constant

Pedagogical judgment, learner empathy, and design thinking remain irreplaceable.

